

Final - Linear Algebra I (2025-26)

Time: 3 hours.

Attempt all questions. The total marks is 52. The maximum score you can obtain is 50.

1. If $\mathbf{H}$ is a matrix in Hermite Canonical Form, show that the non-null columns of $\mathbf{I} - \mathbf{H}$ form a basis of $\mathcal{N}(\mathbf{H})$. [6 marks]
2. Show that the rank of a matrix $\mathbf{A}$ is not altered by pre-multiplying by a nonsingular matrix. [5 marks]
3. Let $\mathbf{A}$ be an $m \times n$ matrix over F . Show that $\mathbf{A}$ has a right inverse if and only if $\mathcal{C}(\mathbf{A}) = F^m$. [6 marks]
4. Let $\mathbf{A}$ be an $n \times n$ matrix. Define $\text{trace}(\mathbf{A})$ to be $\sum_{i=1}^n a_{ii}$, the sum of the diagonal elements. For two matrices $\mathbf{A}, \mathbf{B}$ of dimensions $n \times m$ and $m \times n$ respectively, show that $\text{trace}(\mathbf{AB}) = \text{trace}(\mathbf{BA})$. [5 marks]
5. Let $\mathbf{A}$ be an $m \times n$ matrix. Show that an $n \times m$ matrix $\mathbf{G}$ is a g-inverse of $\mathbf{A}$ if and only if $\rho(\mathbf{I} - \mathbf{GA}) = n - \rho(\mathbf{A})$. [6 marks]
6. Let $\mathbf{x}_1, \dots, \mathbf{x}_k$ be vectors in F^n and let $\mathbf{y}_i$ the vector in F^{n-1} formed by the first $n-1$ components of $\mathbf{x}_i$, $i = 1, 2, \dots, k$. Show that if $\mathbf{y}_1, \dots, \mathbf{y}_k$ are linearly independent in F^{n-1} then $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ are linearly independent in F^n . [6 marks]
7. If $\mathbf{A}$ is an $m \times n$ matrix with rank $r > 0$, show that there exist non-singular matrices $\mathbf{P}$ and $\mathbf{Q}$ of orders m and n respectively such that

$$\mathbf{A} = \mathbf{P} \begin{bmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{Q}.$$

(Hint: row operations and column operations) [8 marks]

8. Solve the system $\mathbf{Ax} = \mathbf{b}$ (find whether it is consistent and a general solution if it is consistent). Also find a g-inverse, the rank and a rank-factorization of $\mathbf{A}$. Here

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 5 \\ 4 & 0 & 8 \\ 3 & 1 & 7 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \quad [10 \text{ marks}]$$